

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PNG 523 — Past Questions & Materials
by Renance

Course: PNG 523

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 1

Levels: 500

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Paper 1 of 1 — PNG 523, 526, 517

Level 500 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY

2017/2018 Second Semester Examination
PNG 523: Production Enhancement & Optimization

Answer All Questions Time: 3 Hours
Please write your Matriculation Number Boldly on your Script

✓1a) Define Completion Interval Potential. Explain how re-perforation can be used to optimize this Potential.

✓1b) The well test data and current production level (CPL) have been found to be very close. Use a flowchart to show how production from this well can be optimized.

2a) Define Intermittent and Continuous Gaslift.

2b) A well was abandoned by a Multi-National Oil Company because it is depleted. However, an Indigenous Company acquired it and put it on intermittent gaslift. The well data are as follows:

Well Depth (Mid Perforations)	=	10,000 ft	
Foot Valve Depth	=	9,000 ft	
Gaslift Valve Depth	=	8,000 ft	
Slug Height	=	3,000 ft	
Oil Gravity	=	0.8	
Gradient of pure water	=	0.433 psi/ft	
Liftgas gradient	=	0.01 psi/ft	
Surface Injection Pressure	=	2500 psig	

b) An alternative due to depth can be coil tubing.

Graphically determine the following:

- The Reservoir Pressure
- The flowing tubing head pressure when the gas arrives at the Gaslift Valve
- The flowing tubing head pressure when the slug arrives at the surface (i.e. the top of the slug is at the surface)

✓3a) What is Beam Pumping? Briefly describe the operation. What are its advantages and disadvantages?

✓3b) If you cannot use Beam Pumping due to its disadvantages in a particular well, what alternative can you use in its place? Why have you chosen this alternative?

a) Beam pumping involves the use of a sucker rod pump (nodding donkey) which is connected to a gas powered motor to pump reservoir fluid from the well bore to the surface.

Adv

- 1) Cheap
- 2) low flow rate ($< 25 \text{ m}^3/\text{d}$)
- 3) can reduce Pwf to zero

Dis

- 1) cannot handle solids, and Gas
- 2) cannot handle high flow rates
- 3) Depth limitation ($< 6000 \text{ ft}$)

- 4a) What is acid washing in well stimulation?
 4b) What is chemical stimulation without acid?
 4c) A well was damaged during drilling and completion. A build-up survey carried out on it confirmed a skin factor of 10. A carefully planned stimulation program was poorly executed and the skin was increased to 15. The reservoir, well and fluid data are shown below. Use the Darcy equation to estimate the pre and post stimulation flow rates for the same flowing bottom hole pressure.

Reservoir Pressure	= 3500 psig	Flowing Bottom Hole Pressure	= 2900 psig
Reservoir Thickness	= 200 ft	Reservoir Permeability	= 300 md
Reservoir Fluid Viscosity	= 35 cp	External Drainage Radius	= 750 ft
Well Bore Radius	= 5.125 inches	Formation Volume Factor	= 1.2 resbbl/stb
API Gravity of Crude	= 30°		

- 5) Management has decided to put a well on continuous gaslift and you have been hired to carry the design given the following specifications:

Valve Type	IPO (Injection Pressure Operated)
Min FWHP	100 psig
Kick-off Pressure	1100 psig
Kill Brine Gradient	0.465 psi/ft
Liftgas Gradient	0.01 psi/ft
Oil with NGLR Gradient	0.360 psi/ft
Well Depth	10,000 ft
Tubing Size	3 1/2"

Below is the Equilibrium Curve

Depth, ft	Pressure, psig
0	200
1000	240
3000	340
4000	450
5000	600
6000	800
7000	1020
8000	1280
9000	1590

- i) Determine the number of mandrels including the orifice.
 ii) What are their depths?
 iii) What is the flowing bottom hole pressure?

$$q = 2\pi k h A$$

- a) Acid washing is a process in well stimulation where major acids (especially hydrochloric acid) are used to remove blockages in the well. These acids melt up the blockages and are flushed together with it.
 b) Chemical stimulation without acid is a process (chemical process) where the well is stimulated using processes such as well fracturing to improve production life.

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING & TECHNOLOGY
 2021/2022 Second Semester Examination
PNG 523: Production Enhancement & Optimization

Answer All Questions **Time: 3 Hours**
Please write your Matriculation Number Boldly on your Script

- 1a) What are low production and off production in Petroleum Production Optimization?
 1b) What is Completion Interval Potential and how do you optimize it?
 1c) Show how well potentials are optimized using a flowchart only.
- 2a) Give reasons for well intervention
 2b) List 6 advantages and 4 disadvantages of wireline (WL).
 2c) A well was damaged during drilling and completion. A build-up survey carried out on it confirmed a skin factor of 25. Acid Stimulation reduced the skin to 3. The reservoir, well and fluid data are shown below. Use the Darcy equation to estimate the pre and post stimulation flow rates for the same flowing bottom hole pressure.

Reservoir Pressure	= 3600 psig	Flowing Bottom Hole Pressure	= 3000 psig
Reservoir Thickness	= 300 ft	Reservoir Permeability	= 400md
Reservoir Fluid Viscosity	= 40 cp	External Drainage Radius	= 750 ft
Well Bore Radius	= 5.125 inches	Formation Volume Factor	= 1.2 resbbl/stb
API Gravity of Crude	= 35°		

- 3a) What is gaslift? Compare and contrast continuous and intermittent gaslift.
- 3b) A well is 10,000 ft deep and produced well fluids of a specific gravity of 0.8 before it quit. The Liquid level was tagged by wire line at 500 ft. There are four mandrels in the well and there are surface gaslift facilities to enable the well to be put on gaslift. The mandrels are located at the following depths: Mandrel No 1 @ 2000 ft; Mandrel No 2 @ 3,000 ft; Mandrel No 3 @ 4000 ft; and Mandrel No 4 @ 5000 ft. Injection of gas through Mandrel No 1 results in a drawdown of 200 psi. Injecting through Mandrel No 2 leads to a drawdown of 400 psi while injecting through Mandrel 3 leads to a drawdown of 600 psi. The orifice valve is located in Mandrel No 4 and injecting through it results in a drawdown of 800 psi. If the aerated liquid has a specific gravity of 0.3 and water gradient is 0.433 psi/ft.

Graphically determine the following:

- a) The reservoir pressure
 b) The flowing bottom hole pressure and flowing tubing head pressure when injecting from Mandrels No 1, No 2, No 3 and No 4.

$$0.8 \times 9500 \times 0.433 \quad 1$$

$$0.8 \times 0.433 \times 8000 \quad 289$$

$$319.6$$

- 4) The Management of an oil company has decided to put a well on continuous gaslift and you have been hired to carry the design given the following specifications:

Valve Type	PPO (Production Pressure Operated)
Min FWHP	100 psig
Kick-off Pressure	1000 psig
Kill Brine Gradient	0.465 psi/ft
Liftgas Gradient	0.01 psi/ft
Oil with NGLR Gradient	0.350 psi/ft
Well Depth	10,000 ft
Tubing Size	3 1/2"

Below is the Equilibrium Curve

Depth, ft	Pressure, psig
0	100
1000	150
3000	340
4000	450
5000	600
6000	800
7000	1020
8000	1280
9000	1590

- Determine the number of mandrels including the orifice.
 - What are their depths?
 - What is the flowing bottom hole pressure?
- 5) If the bellows area and the port area of the valve are 0.31 sq in and 0.029 sq in respectively in your design in Question 4, recommend a setting pressure (TRO) of the valve in each mandrel. The temperature correction factor may be assumed as $1 + 0.000043D$ where D is the valve depth.

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
COLLEGE OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING
SECOND SEMESTER EXAMINATION, 2020/2021 SESSION

PNG 526: RESERVOIR SIMULATION
TIME ALLOWED: 3HRS UNIT: 3

NOTE: ALL CALCULATIONS SHOULD BE REPORTED IN
FOUR DECIMAL PLACE

**SECTION A: ANSWER ANY 3 QUESTIONS FROM
THIS SECTION**

QUESTION 1

a) PVT data derived from laboratory experiments or correlations, are key input parameters for reservoir studies. You are required to generate a table in the format below to list 5 PVT experiments, the type of fluid they can be applied on and three data obtained from each experiment.

Experiment	Type of fluid	Data obtained
1. PVT	Oil	Pressure, Temperature, Volume
2. PVT	Gas	Pressure, Temperature, Volume
3. PVT	Water	Pressure, Temperature, Volume
4. PVT	Oil	Pressure, Temperature, Volume
5. PVT	Gas	Pressure, Temperature, Volume

b) State three equilibration data that are used to establish initial reservoir conditions

QUESTION 2

a) State two parameters each that can be adjusted to achieve a good history match for the following: rock data, fluid data, relative permeability, individual well completion data

b) Make sketches with a label of bubble point pressure, saturated and undersaturated regions of black oil properties such as

- Formation volume factor for oil, gas, & water respectively with one plot
- Viscosity for oil, gas, & water respectively with one plot
- Solution gas-oil ratio.

QUESTION 3

(i) You have been contracted by your company to review data affecting STOIP, if your model value is too high/low, what could be five possible sources of errors?

(ii) State Use a flowchart only, draw a simplified procedure for history matching of:

- Pressure
- Saturation

(iii) If the simulation field pressure is too high, what are the possible changes that could be made to match the history pressure

(iv) In a reservoir simulation model, what are the possible changes that can be made when gas breakthrough is ok, and after breakthrough, the simulation slope is in error

QUESTION 4

- Write the 3D mathematical equation for modeling an Anisotropic reservoir undergoing unsteady state flow.
- State three mathematical model equations that are usually solved in reservoir simulation
- State four deliverables under the following:
 - Seismic review
 - Geologic model
 - BHP Analysis

1

SECTION B: ANSWER ANY 2 QUESTIONS FROM THIS SECTION

QUESTION 5

Show that the diffusivity equation for a slightly compressible fluid is given by:

$$\frac{\partial^2 p}{\partial r^2} + \frac{1}{r} \frac{\partial p}{\partial r} = \frac{\phi \mu C_f}{k} \frac{\partial p}{\partial t}$$

QUESTION 6

The expression below represents a 1D partial differential equation of a single-phase flow reservoir

$$\frac{0.000264k_o \partial^2 p}{\phi \mu_o B_o C_o \partial x^2} + \frac{q_o}{B_o} = \frac{\partial p}{\partial t}$$

If the reservoir is discretized into five grid blocks with the following boundary conditions:

- Characterize the associated matrix fully for all the implicit pressures
- Write down the implicit pressure expression at $i+2$, $n-1$

QUESTION 7

- Use a flowchart only to describe a reservoir simulation model selection in increasing complexity and expense with respect to system, dimension, geometry and phases
- Use a flowchart only, describe the process of reservoir simulation study from problem definition to development plan strategies.

boundary condition
inner boundary (a)
outer boundary (2b)

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY
 Department of Petroleum Engineering
 2019/2020 First Semester

Faith B.

25/01/2021

PNG 517: Integrated Field Development and Planning Mid-semester Test

Time Allowed 30mins

Question

2. FUPRE Oil & Gas Nigeria limited is to develop a new field offshore Niger Delta for maximum production using the schematic in Fig. 1 below. The production system design was sub-contracted to three indigenous service companies with different proposed production systems as shown in the Table 1 below. With maximum production as the overriding consideration, which of the proposed systems would you implement for the field if you are the production manager?

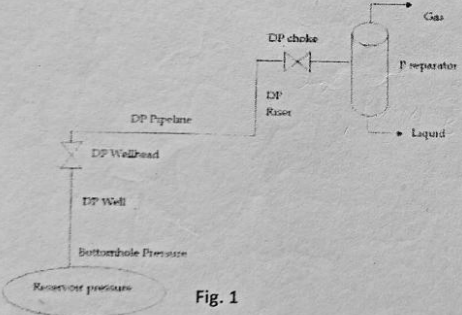


Fig. 1

Table 1

Pressure (Bar)	Lamb Ltd.	Ken E&P	Post-HND Energy
P separator	20	30	70
DP choke	5	2	10
DP riser	2	5	1
DP pipeline	3	1	5
DP well head	20	40	50
DP well	3	5	10
	Σ	Σ	Σ

↑ highest summation ✓

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



COLLEGE OF ENGINEERING AND TECHNOLOGY

Department of Petroleum Engineering

2020/2021 First Semester Examination

PNG 517: Integrated Field Development and Planning

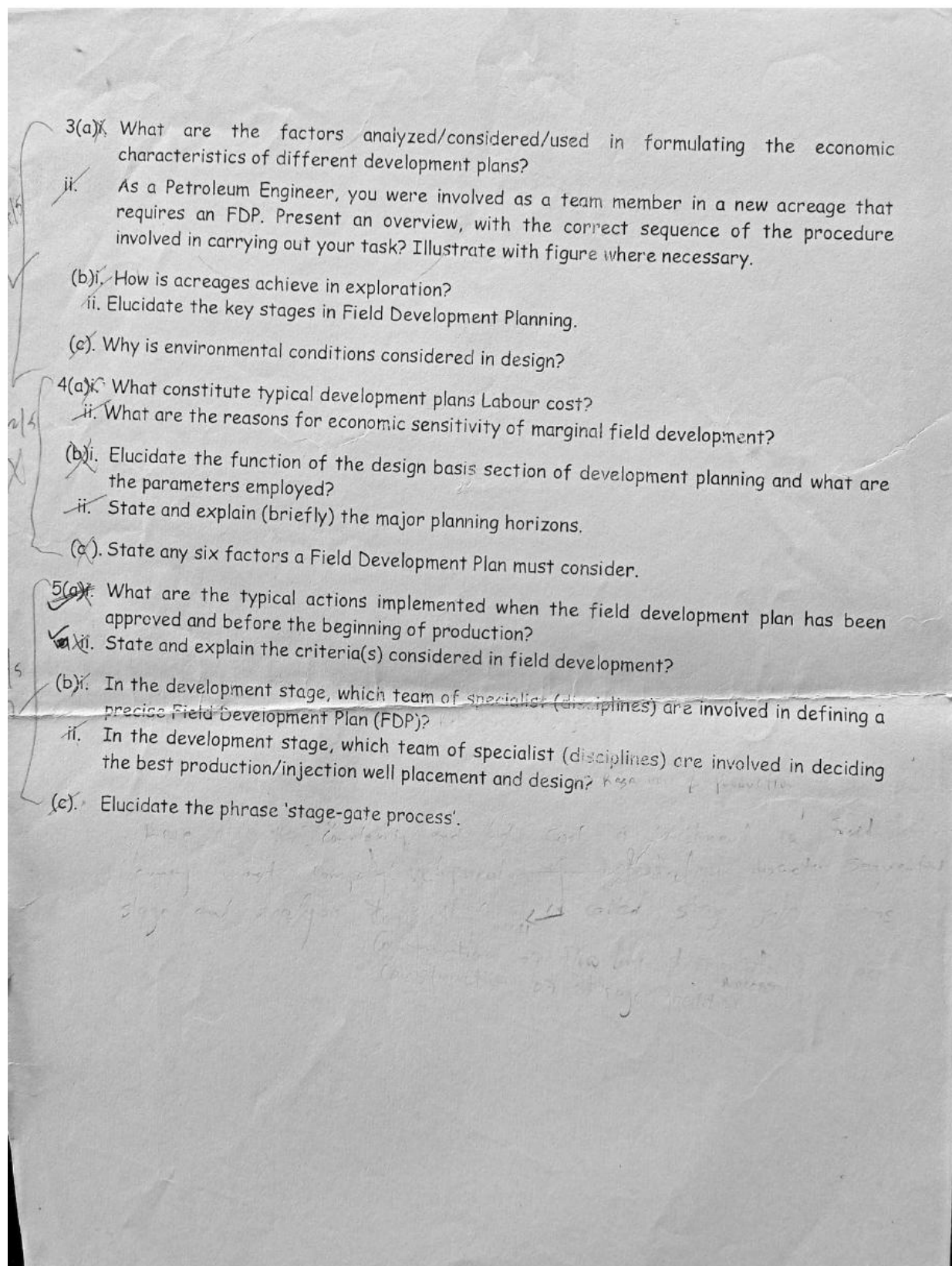
Answer Question 1 and any other 3.

Time Allowed: 3 Hrs.

22/11/2021

QUESTIONS

- 1(a)i. What happens when an exploration well has been fully approved and the drilling activity is being prepared?
- ii. The development of an oil and gas field costs millions of dollars and may require long time (5-10 years) to be fully realized. What are the factors that can influence the costs and duration of the development phase? *new gas field*
- (b)i. As a Trainee Fluid Specialist with the reservoir engineering team of an Operating company, what will consider as the three relevance of phase behaviour study to the success of the company's new field development project?
- ii. They are different methods adopted in hydrocarbon volume estimations which can be grouped into the deterministic or probabilistic approaches. Which of these will provide a more realistic result and explain why?
- (c)i. Calculate the initial oil in place for an Oil Field with the following given data:
 N = Oil in place (bbl);
 Net productive thickness = 49ft; *Area = 26,700 acres*
 Initial reservoir pressure, p_i = 2980 psia; *Porosity = 8%*
 B_o at p_i = 1.68 bbl/STB
 B_o at p_a = 1.15 bbl/STB; *S_g at p_a = 34%*
 S_{or} after water invasion = 20%; *Average S_w = 45%*
Handwritten notes: $N = N_r \frac{V_p (1 - S_{w_i})}{B_o}$, $S_{or} = 1 - S_w + S_g$
- ii. During the life of a field, the reservoir simulation study results will be dependent on which parameters?
- 2(a)i. Why will you recommend to have a wide spacing at the beginning when planning a reservoir development system?
- ii. An investor has invested more capital in an on-going maintenance operation, would you consider this to be a project? Explain the reason behind your consideration.
- (b)i. Today, why does a typical field development planning involves greater detailed studies? Mention at least 4 points.
- ii. What is Field Development Planning?
- (c). Explain the engineering section of development plan and mention the key elements.



Faith B.

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



COLLEGE OF ENGINEERING AND TECHNOLOGY

Department of Petroleum Engineering

2019/2020 First Semester Examination

PNG 517: Integrated Field Development and Planning

Answer Question 1 & any other three

Time Allowed: 3hrs. Date: 15/03/2021

QUESTIONS

- 1(a)i. As a Trainee Fluid Specialist with the reservoir engineering team attached to the Fluid Services unit of FUPRE E&P Ltd. Highlight the relevance of phase behaviour study to the success of FUPRE E&P new field development project.
- ii. Describe three classifications of conventional gas reserves based on the condition of the gas in the reservoir?
- (b)i. The development of an oil and gas field costs millions of dollars and may require long time (5-10 years) to be fully realized. What are the factors that can influence the costs and duration of the development phase?
- ii. State and explain typical production system options for offshore field development?
- (c)i. Using the information provided below, calculate the oil resource in place for a conventional reservoir.

$$N = VR \cdot \phi \cdot 1 / B_o \cdot (1 - S_w)$$

N= Oil in place (bbl);

Average pay thickness = 25ft;

Formation volume factor = 0.9 rb/stb;

Area of the reservoir = 30ft²;

Porosity = 8%

Water Saturation = 15%

- ii. State and explain (briefly), the stage that takes place after successfully completing the appraisal period and before the beginning of the field production?
- 2(a)i. Mention at least three information or focal points that must be defined and economically evaluated in a typical petroleum field development project.
- ii. Briefly explain the key stages of a FDP project.
- (b)i. Today, why does a typical field development planning involves greater detailed studies? Mention at least 4 points.
- ii. How do we achieve acreages?
- (c). Elucidate the phrase 'stage-gate process'.

Handwritten calculations:

1 acre = 4850 ft²

7758 acre-ft/bbl

1 ft = 0.3048 m

1 ft² = 0.09290304 m²

30 ft² = x

4.645152 m²

- 3(a)i. What stage in FDP entails the long-term management of production operations and what is the objective of the stage?
- ii. As a young Petroleum Engineer, you were involved as a team member in a new acreage that requires an FDP. Present an overview, with the correct sequence of the procedure involved in carrying out your task? Illustrate with figure where necessary.
- (b)i. What is FDP?
- ii. An investor has invested more capital in an on-going maintenance operation, would you consider this to be a project? Explain the reason behind your consideration.
- (c). Distinguish between stages and sections in FDP using appropriate examples where necessary?
- 4(a)i. Explain marginal field development and state the reasons for economic sensitivity of marginal field development?
- ii. State at least five (5) key elements of the exploitation section.
- (b)i. Elucidate the function of the design basis section of development planning and what are the parameters employed?
- ii. State some (at least 7) important components of the design basis section.
- (c). Use a figure to illustrate a field development cycle, and state any six factors that can affect options employed in field development?
- 5(a)i. You were requested to do a SWOT analysis of a field development. Describe how you will embark on the task (Use an example of a field development scenario)?
- ii. State and explain the criteria(s) considered in field development?
- (b)i. A team of the required specialist executed a field development. State the key indicators they will have to use to define a successful project and also the hindrance to such?
- ii. In question (bi), what methods would you suggest to the team to improve the performance of developed field?
- (c). Explain the engineering section of development plan and mention the key elements.

$$XPS = \frac{141.5 - 131.5}{8}$$

$$AP(131.5) = \frac{141.5}{8}$$

$$8 = \frac{141.5}{AP(131.5)}$$